Nine Education IIT Academy

Class 11 — Physics Test Paper

SET B

Name: _______________________ **Roll No.:** ___________ **Class/Section:** _______________ **Date:** _______________

Total Marks: 30

Instructions: Answer all questions. Show all working clearly.

Section 1 — 2 Marks Questions *(5 × 2 = 10 Marks)*

1. Evaluate $\cos 210°$ and $\tan 315°$.
2. Write the $\tan 2\theta$ formula.
3. Find the magnitude of the vector $2\hat{i} - 3\hat{j} + 4\hat{k}$.
4. Find **A** × **B**, where **A** = $\hat{i} - 2\hat{j} + 4\hat{k}$ and **B** = $2\hat{i} - \hat{j} + 2\hat{k}$.
5. If $y = 3x^2 + \sin x + 2\log_e x$, find dy/dx.

Section 2 — 4 Marks Questions *(3 × 4 = 12 Marks)*

1. **A** = $2\hat{i} + 3\hat{j} + 4\hat{k}$, **B** = $3\hat{i} + 4\hat{j} + 5\hat{k}$. (a) Find the resultant of **A** and **B.** (b) Find the angle between **A** and **B.** (c) Find the projection of **A** on **B.**
2. Using a screw gauge, the observations of the diameter of a wire are 1.324, 1.326, 1.334, and 1.336 cm respectively. Find the average diameter, the mean error, the relative error, and the percentage error.

3. The position of a body is given by $x = At + 4Bt^3$, where A and B are constants, x is the position, and t is time. Find: (a) acceleration as a function of time (b) velocity and acceleration at $t = 5$ s

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Section 3 — 8 Marks Question *(1 × 8 = 8 Marks)*

1. If $\mathbf{a}$ and $\mathbf{b}$ make an angle $\cos^{-1}(5/9)$ with each other, such that $|\mathbf{a} + \mathbf{b}| = \sqrt{2}\,|\mathbf{a} - \mathbf{b}|$, and $|\mathbf{a}| = n|\mathbf{b}|$, find the value of n.

All the best!